

## 1.1.2.2 – Features of the Operating Systems

**Q1** Computers usually have operating systems.

Describe what an operating system is. (3)

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**Q2** A user sends a document to their printer

a) Describe the stages of what happens to the document that's sent (2)

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b) State the name of what this process is called (1)

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c) State an advantage of this process (1)

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**Q3** There may be multiple accounts on an operating system.

State what security control allows (2)

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**Q4** Operating systems tend to have back-up space

a) How does an operating system manage back-up space? (1)

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b) State the name of a filing system. You may use an acronym. (1)

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c) State the structure files are organised in. (1)

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**Q5** Multiple processes can happen at once on a computer.

How does the operating system manage these processes? (3)

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**Q6** Compression is the process of making a file smaller.

a) Explain why this is an advantage. (2)

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b) Compare Software & Hardware based disk compression. (5)

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**Q7** Files are stored on computer systems.

a) Describe disk fragmentation and describe the process of defragmentation. (4)

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b) Explain why disk fragmentation takes longer to access files. (2)

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**Q8** Most computers have user interfaces.

State 4 Examples of how the operating system provides a user interface. (4)

- 1) 

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- 2) 

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- 3) 

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- 4) 

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## MARK SCHEME

### Main guidance:

Words in **bold** must be included within answer

Words in (brackets) don't have to be included

Words in {curly brackets} have multiple correct answers. E.g. {FAT/NTFS}

| Question                                                                                                                                                                                                                              | Answer                                                                                                                                                                                                                                                                                                                       | Additional Guidance |
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| <b>Q1</b> Computers usually have operating systems. Describe What an operating system is. (3)                                                                                                                                         | <ul style="list-style-type: none"> <li>• It's <b>software</b> that <b>manages</b> a <b>computer system</b></li> <li>• It's loaded by the {<b>bootstrap</b> (loader)/<b>BIOS</b>}</li> <li>• Primary function is to <b>manage resources</b></li> <li>• Another function is to provide a {<b>UI/User interface</b>}</li> </ul> | Max 3 marks         |
| <b>Q2</b> A user sends a document to their printer<br>a) Describe the stages of what happens to the document that's sent (2)<br><br>b) State the name of what this process is called (1)<br>c) State an advantage of this process (1) | <ul style="list-style-type: none"> <li>• Data's <b>stored</b> in {a queue/hard disk/memory}</li> <li>• It's printed when the printer is {<b>free/in correct order</b>}</li> <li>• <b>Spooling</b></li> <li>• User can {<b>carry on working/log off/do other things</b>} (when waiting for it to print)</li> </ul>            |                     |
| <b>Q3</b> There may be multiple accounts on an operating system. State what security control allows (2)                                                                                                                               | <ul style="list-style-type: none"> <li>• The <b>creation</b> and <b>deletion</b> of (user) accounts</li> <li>• {<b>Change/edit</b>} passwords</li> <li>• Logon</li> </ul>                                                                                                                                                    | Max 2 marks         |
| <b>Q4</b> Operating systems tend to have back-up space<br>a) How does an operating system manage back-up space? (1)<br><br>b) State the name of a filing system. You may use an acronym. (1)                                          | <ul style="list-style-type: none"> <li>• (Ensures that) data can be <b>retrieved</b> from <b>any</b> disk drive</li> <li>• {<b>FAT/NTSF</b>}</li> <li>• {<b>File allocation Table/New Technology File Systems</b>}</li> </ul>                                                                                                | Max 1 mark          |

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| c) State the structure files are organised in. (1)                                                                                                                                | <ul style="list-style-type: none"> <li>• <b>Hierarchical</b> (directory)</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                     |
| <b>Q5</b> Multiple processes can happen at once on a computer.<br>How does the OS manage these processes? (3)                                                                     | <ul style="list-style-type: none"> <li>• On a multitasking OS lets all tasks run <b>{simultaneously/at the same time}</b></li> <li>• (Ensures that) different processes can use the CPU</li> <li>• Programs don't <b>{interfere with each other/crash}</b></li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                     |
| <b>Q6</b> Compression is the process of making a file smaller.<br>a) Explain why this is an advantage. (2)<br><br>b) Compare Software & Hardware based disk compression. (6)      | <ul style="list-style-type: none"> <li>• Allows <b>more</b> data to be stored (on the disk)</li> <li>• Files can be transferred <b>faster</b></li> </ul> <p><b>General:</b> (×1 Mark)</p> <ul style="list-style-type: none"> <li>• Disk based compression is <b>lossless</b></li> </ul> <p><b>Inclusion:</b> (×3 Marks)</p> <ul style="list-style-type: none"> <li>• Software based is often <b>included in operating systems...</b></li> <li>• But, hardware based <b>requires specialist hardware...</b></li> <li>• Specialist hardware may be <b>expensive</b></li> </ul> <p><b>Speed:</b> (×1 Marks)</p> <ul style="list-style-type: none"> <li>• Software based compression is <b>slower</b> than hardware based compression.</li> </ul> | Allow alternative wording that means the same thing. E.g. synonyms. |
| <b>Q7</b> Files are stored on computer systems.<br>a) Describe disk fragmentation and describe the process of defragmentation. (4)<br><br>b) Explain why disk fragmentation takes | <ul style="list-style-type: none"> <li>• Disk fragmentation happens over time</li> <li>• Disk fragmentation is when parts of a file are stored on different parts of the disk</li> <li>• Defragmentation is when files are physically re-arranged on a disk</li> <li>• Meaning that the file are stored next to each other</li> </ul><br><ul style="list-style-type: none"> <li>• Files are stored at different parts of the disk</li> </ul>                                                                                                                                                                                                                                                                                                  |                                                                     |

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| longer to access files. (2)                                                                                                       | <ul style="list-style-type: none"> <li>• So, the read-write head doesn't need to keep rotating the disk that takes time</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                           |                                               |
| <p><b>Q8</b> Most computers have user interfaces. State 4 Examples of how the operating system provides a user interface. (4)</p> | <ul style="list-style-type: none"> <li>• Allows {<b>copying/deleting/moving/sorting/searching</b>} of {<b>files/folders</b>}</li> <li>• Allows access to system settings (such as hardware)</li> <li>• Provides a {<b>command line interface/CLI</b>}</li> <li>• Allows users to have more than one window open</li> <li>• Provides a {<b>graphical user interface/GUI</b>}</li> <li>• Provides user with error messages</li> <li>• Allows for customising the interface</li> <li>• Allows user to switch between {<b>tasks/programs/windows</b>}</li> </ul> | <p>Max 4 marks.</p> <p>No repeat answers.</p> |